

Effective Communication in Cheap-Talk Games^{*}

Sidarta Gordon[†] Navin Kartik[‡] Melody Lo[§]

Wojciech Olszewski[¶] Joel Sobel^{||}

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Abstract

This paper presents equilibrium selection arguments based on learning and dominance in the [Crawford and Sobel \(1982\)](#) model of cheap talk. Our starting point is to restrict players to monotonic strategies with respect to an exogenous ordering of messages. That by itself does not alter equilibrium outcomes. However, under a standard regularity condition, robust best-response dynamics from arbitrary initial conditions converges to a unique equilibrium: the “most informative” equilibrium with “maximally exaggerated” language. We also offer a process of iterated deletion of weakly dominated strategies that selects the same equilibrium.

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[†]LEDA, Université Paris-Dauphine, email: sidarta.gordon@dauphine.psl.eu

[‡]Department of Economics, Yale University, email: nkartik@gmail.com

[§]Department of Economics, National Taiwan University, email: peiyulo2006@gmail.com

[¶]Department of Economics, Northwestern University, email: wo@kellogg.northwestern.edu

^{||}Department of Economics, University of California, San Diego, email: jsobel@ucsd.edu

1 Introduction

Talk is a useful way to communicate private information in strategic situations, as formalized by Crawford and Sobel (1982) and Green and Stokey (2007). However, these—and other—models of cheap talk typically have multiple equilibrium outcomes, including an uninformative one in which no information is transmitted. A central concern in the literature has been finding conditions under which communication is *effective*, that is in which the predicted outcome involves non-trivial information transmission.

This paper contributes to that effort. We develop arguments that lead to the selection of equilibria with effective communication in the Crawford and Sobel (1982) (hereafter CS) model. We present two sets of arguments, one based on learning (best-response dynamics) and one based on deductive reasoning (iterated dominance). Although the approaches are distinct, there is a common thread. In either case, we obtain the same selection of not only the payoff-relevant outcome, but also the use of language.

Recall that the CS model has an informed Sender communicating to an uninformed Receiver. The Receiver responds to the Sender’s message by making a decision that is payoff relevant to both players. Talk is cheap because neither party’s payoffs depend directly on the message. CS characterize the set of equilibrium outcomes in a one-dimensional environment with an “upward-bias” conflict of interest: the Sender always prefers higher decisions than the Receiver. There is a finite upper bound, N^* , on the number of distinct actions that the Receiver takes in equilibrium; for each $N = 1, \dots, N^*$, there is an equilibrium in which the Receiver takes N distinct actions. In addition, under a regularity condition (“Condition M” in CS), the equilibrium outcome for each such N is unique, and the ex-ante expected payoff for both Sender and Receiver is strictly increasing in N . The outcome with N^* actions is typically what applications focus on.

The multiple-equilibria problem arises in three different ways in cheap-talk games. First, some messages may not be used in equilibrium, and there will typically be many specifications of the Receiver’s off-path behavior that support the equilibrium. This kind of off-path indeterminacy is familiar in games with incomplete information. The second kind of multiplicity, message indeterminacy, is about which messages are used in equilibrium and their meaning. Given any equilibrium, one can generate another equilibrium by permuting the interpretation of messages. This reflects an arbitrariness of language. It would be of limited concern if all equilibria induce the same mapping from Sender types to Receiver actions.¹

¹The word used to describe the color of a white house in Paris is “*blanche*” and in Warsaw is “*biały*”. What matters is that French speakers and Polish speakers classify the same set of houses as white (and their

However, there is a more problematic kind of multiplicity, outcome or type-action indeterminacy. Cheap-talk games invariably have an uninformative equilibrium,² but often also have informative equilibria in which the Receiver takes at least two different actions with positive probability. In the CS model, this arises when the conflict of interest is not too large, and the aforementioned upper bound N^* is at least 2. Indeed, as N^* grows unboundedly as conflict vanishes, the outcome multiplicity can be severe. It is this kind of multiplicity that we are primarily concerned with. Our approach to addressing it, however, starts from a hypothesis that mitigates the degree of message indeterminacy.

Formally, we augment the CS model with a total order on the message space (taken to be finite, of size at least N^*) and restrict players to strategies that are monotonic with respect to that order. This defines a *monotonic cheap-talk game*. We view this formulation as a way to incorporate exogenous meaning or conventions into communication: players enter the strategic setting with a shared ordering of messages and it is common knowledge that they will behave in a way that is consistent with this ordering. Monotonic strategies rule out certain kinds of message indeterminacy: it allows for grades B and A to respectively mean “average” and “above average” in one equilibrium, while they could respectively mean “above average” and “excellent” in another equilibrium; but B can never mean “above average” when A means “average”.³

The monotonic cheap-talk game does not by itself alleviate outcome (i.e., type-action) multiplicity: all CS equilibrium outcomes are preserved. However, classic approaches to equilibrium selection, which have no bite absent the monotonicity restriction, are now fruitful.

Our first approach is based on best-response dynamics. Starting from an arbitrary pair of monotonic Sender and Receiver strategies, we study the iteration of *robust* best responses for each player. Robustness handles indifference: how a player chooses between multiple best responses stemming from either off-path messages (for the Receiver) or on-path multiplicity (for the Sender). Roughly, a strategy x is a robust best response to some opponent strategy if all small perturbations of the opponent’s strategy have best responses close to x . Each player has a unique (up to a measure-zero qualification) robust best response to any opponent strategy. We establish in [Proposition 1](#) that under the CS regularity condition, the iteration of robust best responses from arbitrary initial conditions in the monotonic-cheap talk game

audiences understand that) rather than the particular word used to describe the color.

²To be precise, they typically have many uninformative equilibria when one takes into account the first two kinds of multiplicity.

³We do not suggest, of course, that nonmonotonic strategies are never relevant; consider irony or sarcasm. We are concerned with settings in which players do not entertain such language use—grading being one example.

converges to a unique strategy profile, which is an equilibrium. This equilibrium uses only the highest N^* messages, and its outcome is the N^* -action outcome of CS.

Our second approach is based on iterated dominance. Throughout, by “dominance” we mean “weak dominance”. Since the cheap-talk game is a sequential-move Bayesian game, we view the appropriate notion of dominance to be *interim* dominance: the standard notion of dominance must hold for each type (up to a measure-zero qualification) of the Sender; and on the Receiver side, for each message that is received with positive probability.⁴ We study a process of iterated deletion of interim weakly dominated strategies (IDIWDS) in the monotonic cheap-talk game. This process is related to robust best response iteration, but iterative deletions are justified by iterative applications of interim dominance. **Proposition 2** establishes that, under the CS regularity condition, our process of IDIWDS selects a unique strategy profile, which is an equilibrium—the same equilibrium as selected by robust best-response iteration. Since interim dominance implies dominance (and because there is a unique surviving strategy profile), this process also yield the same selection through iterated deletion of weakly dominated strategies.

The upshot is that both our selection results identify not only the N^* -action outcome, but an equilibrium in which that is induced via the highest N^* messages. In other words, there is maximal exaggeration or inflated language, which we find intuitive in light of the Sender’s upward bias.

The proofs for both our approaches operate via an analysis of two sequences of strategy profiles. The sequences are defined by specifying extreme initial conditions—one “highest” and one “lowest”—and then iterating robust best responses. The highest (resp., lowest) initial condition is that the Receiver take the highest (resp., lowest) rationalizable action in response to every message and the Sender always sends the lowest (resp., highest) message. We show that the sequences defined by iteration from these conditions are suitably monotone—the highest sequence decreases while the lowest sequence increases—and they both converge to equilibria. We then appeal to **Chen, Kartik and Sobel’s (2008)** no-incentive-to-separate (NITS) condition on equilibria. They showed that any N^* -action outcome always satisfies NITS, and it is the unique NITS outcome under CS’ regularity condition. We establish

⁴So a Sender strategy s interim weakly dominates s' against a set of Receiver strategies if all (except perhaps a set of measure zero) Sender types weakly prefer s to s' against every Receiver strategy under consideration, and a positive measure of types strictly prefer s to s' against some Receiver strategy. A Receiver strategy a interim weakly dominates a' against a set of Sender strategies if the Receiver weakly prefers a to a' no matter which positive-probability message is sent by any of the Sender strategies, and there is some positive-probability message under some Sender strategy such that the Receiver strictly prefers a to a' .

that so long as there is a unique NITS outcome, our two sequences have a common limit (Theorem 1/Proposition 1). This limit equilibrium satisfies NITS and hence induces N^* actions. Moreover, it is the highest N^* messages that are used in this equilibrium; all lower off-path messages would elicit the lowest rationalizable action, in line with NITS’ defining property. We thus provide a foundation for the NITS condition. Our results on robust best-response iteration follow because any sequence of robust best responses, starting from an arbitrary initial condition, must be sandwiched between the highest and lowest sequences. Our results on IDIWDS follow because we can show that strategies larger than the higher limit or lower than the lower limit must eventually be deleted by our process. The exact process of iterated deletion involves nuances to deal with unused messages, but roughly speaking involves deleting in each round (on the basis of interim dominance) all strategies that are more extreme than the highest and lowest sequences’ strategies.

We note that the restriction to monotonic strategies plays different roles in our two approaches to selection. For iterated dominance, monotonicity is essential; without it, the dominance arguments have no traction. For best-response dynamics, however, the restriction is convenient but not essential: any robust best-response sequence in the original game can be mapped via message relabeling to a monotonic sequence, so our convergence result applies to the unrestricted game as well; see Remark 1.

As mentioned at the outset, there has been substantial interest in refining predictions in cheap-talk games, including specifically for the CS model. We defer a discussion of the literature to Section 6. Here we just note that each element of our approach has antecedents. In particular, Chen, Kartik and Sobel (2008) also appeal to monotonic strategies (as an equilibrium refinement in perturbed games with lying costs or nonstrategic types) in the CS model, whereas best response and dominance iteration are classical game-theoretic refinements. What is novel, and we think somewhat surprising, is how powerful these classical approaches—in particular, iterated dominance—become in the CS model under the restriction to monotonic strategies. But cheap talk introduces some subtle issues in the iteration arguments stemming from indifferences. Hence we appeal to robust best response iteration, and also offer only a specific (although broadly intuitively) order of elimination for IDIWDS. We do not prove that the outcome is order independent, although we conjecture that it is.

Outline. The remainder of the introduction presents a simple example to illustrate ideas. Section 2 introduces the monotonic cheap-talk game we study and some basic properties of equilibria. Section 3 develops our key concepts of robust best responses and dominance. Section 4 shows how our iterative arguments work in a uniform-quadratic two-message exam-

ple, providing intuition for the formal arguments. [Section 5](#) then presents the main results. [Section 6](#) discusses related literature. Omitted proofs are in the appendices.

1.1 A Common-Interest Example

We wrap up this introduction by analyzing a common-interest cheap-talk game. The example does not fit into our main model, but it provides the simplest illustration of how monotonicity in conjunction with dominance/learning arguments can yield selection.

There are two equiprobable Sender types (high and low), two Receiver actions (also high and low, H and L), and two messages the Sender chooses from (also high and low, h and l). Both Sender and Receiver receive a payoff of two if the action matches the type and a payoff of zero otherwise. A pure strategy for the Sender strategy is a pair (i, j) where the Sender sends message i when her type is low and j when her type is high. Similarly, a Receiver pure strategy is a pair (i, j) where i is the action taken after message low and j after message high. The strategic form of the game is given in the following table, where rows correspond to Sender strategies and columns to the Receiver.

	(H, H)	(L, H)	(H, L)	(L, L)
(h, h)	1, 1	1, 1	1, 1	1, 1
(h, l)	1, 1	0, 0	2, 2	1, 1
(l, h)	1, 1	2, 2	0, 0	1, 1
(l, l)	1, 1	1, 1	1, 1	1, 1

There are two efficient pure-strategy equilibria, $((l, h), (H, L))$ and $((h, l), (H, L))$, in which the Sender distinguishes between the states and the Receiver correctly interprets this information. The former is more intuitive than the latter, as the latter flips the natural association of types with messages and messages with actions. But the game also has an uninformative and ex-ante Pareto inefficient equilibrium in which the Sender mixes uniformly between (h, l) and (l, h) and the Receiver mixes uniformly between (H, L) and (L, H) .⁵ The mixed-strategy equilibrium, as well as both efficient pure-strategy equilibria, satisfy standard refinements from perfection to strategic stability.

Our approach is to replace the original game by a game in which non-monotonic strategies are not available. Monotonicity here is with respect to the natural ordering on messages ($l < h$), types (low < high), and actions ($L < H$). The strategic form of the monotonic

⁵ There are also other inefficient equilibria, in pure and mixed strategies.

cheap-talk game is:

	(H, H)	(L, H)	(L, L)
(h, h)	1, 1	1, 1	1, 1
(l, h)	1, 1	2, 2	1, 1
(l, l)	1, 1	1, 1	1, 1

Deleting non-monotonic strategies has eliminated some inefficient equilibria but it does not eliminate any equilibrium outcome, i.e., equilibrium mapping from types to (distributions over) actions. In particular, the previous mixed-strategy equilibrium outcome is replicated in an equilibrium where the Sender plays (l, l) (or (h, h)) and the Receiver mixes uniformly over (L, L) and (H, H) . However, weak dominance now selects the $((l, h), (L, H))$ equilibrium, which is efficient. Note that weak dominance has no power in the original game. It is the combination of monotonic strategies—which eliminates certain (but not all) kinds of message indeterminacy—and an equilibrium refinement that yields selection.

Instead of applying weak dominance in the monotonic game, we can also obtain the same selection through a version of best-response dynamics starting from any initial condition. For an arbitrary initial condition—in particular, an inefficient pure-strategy equilibrium—standard best-response dynamics are clearly not sufficient. Our approach is to require *robust* best responses: a best response such that there is a “nearby” best response to any “nearby” opponent strategy. Because of weakly dominant strategies in this example, it is straightforward that the unique robust best response to any Sender strategy is (L, H) and similarly the unique robust best response to any Receiver strategy is (l, h) . So one iteration of robust best responses converges to the $((l, h), (L, H))$ efficient equilibrium. Again, robust best response iteration would not provide selection in the original game.⁶

Balkenborg, Hofbauer and Kuzmics (2015, Section 6) and Myerson and Weibull (2015, Example 6) use the same example to illustrate the power of other refinement arguments. Both of these papers also select an efficient outcome. Balkenborg, Hofbauer and Kuzmics eliminate the Sender’s strategies (h, h) and (l, l) as “weakly inferior” (they are equivalent to an equal mixture of the other two strategies) and point out the efficient outcomes are the only locally stable equilibria of a refined best-reply dynamic that avoids these strategies.

⁶In the original game, robust best-response dynamics are not even well-defined. In particular, the Receiver has no robust best response there to the Sender strategy (h, h) : intuitively, any nondegenerate mixture over (h, h) and (l, h) has a unique best response of (L, H) , while any nondegenerate mixture over (h, h) and (h, l) has a unique best response of (H, L) . Thus no single Receiver strategy has a “nearby” best response to each Sender strategy “near” (h, h) .

Myerson and Weibull show that only the efficient equilibria are settled. Settled equilibria are a refinement of the set of proper equilibria (Kalai and Samet, 1984).

2 Model

2.1 The Cheap-Talk Game

We start from the CS model (Crawford and Sobel, 1982). There are two players. A Sender privately observes his type t drawn from a continuous density $f > 0$ on $[0, 1]$. The Sender then sends a message m from a set M , where M is large enough (elaborated below). The Receiver observes m and chooses an action $a \in \mathbb{R}$. Payoffs are $u^S(a, t)$ for the Sender and $u^R(a, t)$ for the Receiver. For each player $i = S, R$, the payoff function u^i is twice continuously differentiable, strictly concave in a ($u_{aa}^i < 0$), and strictly supermodular ($u_{at}^i > 0$), where subscripts denoted partial derivatives as usual. Each player i has a type-dependent ideal action, $a^i(t) := \arg \max_a u^i(a, t)$.

Assume the Sender has an *upward bias*: $a^S(t) > a^R(t)$ for all $t \in [0, 1]$. We normalize $a^R(0) := 0$ and $a^R(1) := 1$; this implies that we can restrict attention to actions in $[0, 1]$, as other actions are strictly dominated for the Receiver.⁷

2.2 The Structure of Equilibria

A pure strategy for the Sender is a mapping $s : [0, 1] \rightarrow M$, while a pure strategy for the Receiver is a mapping $a : M \rightarrow \mathbb{R}$. The notion of equilibrium is Bayes-Nash. An equilibrium *outcome* is the mapping from types to (distributions over) actions.

CS demonstrate that for equilibrium outcomes (and up to measure zero sets of types) it is without loss to consider only pure strategy equilibria. There is a positive integer N^* such that for every $n \in \{1, \dots, N^*\}$, there is an equilibrium in which there are n induced actions (i.e., actions played with ex-ante positive probability); moreover, there is no equilibrium that induces strictly more than N^* actions. For any $0 \leq t' < t'' < 1$, let

$$a^R(t', t'') := \arg \max_a \int_t^{t''} u^R(a, t) f(t) dt \quad (1)$$

be the Receiver's optimal action when she only learns that the Sender's type lies in $[t', t'']$. Any equilibrium can be characterized by cutoffs $0 = t_0 < t_1 < \dots < t_n = 1$, and actions

⁷The sender's ideal action exceeds 1 for some types, but for those types the most preferred "feasible" action is 1.

$0 < a_1 \leq \dots \leq a_n < 1$ such that

$$u^S(a_{i+1}, t_i) - u^S(a_i, t_i) = 0 \quad (2)$$

for $i = 1, \dots, n-1$, and

$$a_i = a^R(t_{i-1}, t_i) \quad (3)$$

for $i = 1, \dots, n$. Any equilibrium has n distinct messages played with positive probability and types in (t_i, t_{i+1}) pooling on a common message. Condition (2) states that the cutoff types are indifferent between pooling with types immediately below or immediately above. Condition (3) states that the Receiver best responds to information in the Sender's message. Ranging over $n \in \{1, \dots, N^*\}$, conditions (2) and (3) fully characterize all the equilibrium outcomes (up to measure zero sets, stemming from the Sender's behavior at the cutoffs; we ignore measure-zero qualifications hereafter).

In general, there can be multiple equilibrium outcomes for a given $n \in \{2, \dots, N^*\}$. CS introduce a technical regularity condition (Condition “(M)” in their paper) that guarantees uniqueness for each n .⁸ The condition is satisfied, by the leading “uniform-quadratic” example in CS, which has been the focus of many applications.

U-Q Specification. The prior density $f(t) = 1$ on $[0, 1]$, the Receiver's utility is $u^R(a, t) = -(a - t)^2$ and the Sender's utility is $u^S(a, t) = -(a - t - b)^2$ for some bias parameter $b > 0$.

CS prove that when the regularity condition holds, then not only is there is a unique equilibrium outcome for each $n \in \{1, \dots, N^*\}$, but moreover, the ex-ante equilibrium expected utility for both the Sender and Receiver is strictly increasing in N . That provides one argument for the salience of the N^* equilibrium outcome.

Our analysis will use the following condition from [Chen, Kartik and Sobel \(2008\)](#).

Definition 1. An equilibrium (s^*, a^*) satisfies *No Incentive to Separate* (NITS) if

$$u^S(a^*(s^*(0)), 0) \geq u^S(a^R(0), 0).$$

⁸For completeness, we restate their condition here. For $t_{i-1} \leq t_i \leq t_{i+1}$, let

$$V(t_{i-1}, t_i, t_{i+1}) := u^S(a^R(t_i, t_{i+1}), t_i) - u^S(a^R(t_{i-1}, t_i), t_i).$$

A (forward) solution to (2) of length L is a sequence $t_0 < \dots < t_L$ such that $V(t_{i-1}, t_i, t_{i+1}) = 0$ for $i \in \{1, \dots, L-1\}$ and $t_0 < t_1$. CS' regularity condition requires that for any two solutions to (2) of length L , $(t_0, \dots, t_L)$ and $(t'_0, \dots, t'_L)$ with $t_0 = t'_0$ and $t_1 < t'_1$, it holds that $t_i < t'_i$ for all $i \geq 2$.

NITS states that the lowest type of the Sender prefers her equilibrium payoff to the payoff she would receive if the Receiver knew her type (and responded optimally). [Chen, Kartik and Sobel \(2008\)](#) show that every equilibrium with N^* induced actions satisfies NITS, and that under CS' regularity condition, only the unique equilibrium outcome with N^* actions satisfies NITS.

2.3 The Monotonic Cheap-Talk Game

So far, the message space has been abstract. From now on, we assume it is finite and ordered: $M := \{m_1, \dots, m_N\}$, where N is a positive integer and $m_i < m_{i+1}$ for each $i \in \{1, \dots, N-1\}$. Here “ $<$ ” denotes the order on M ; one can take $M \subset \mathbb{R}$. We are interested in settings with $N \geq N^*$, where N^* is the upper bound on equilibrium actions described in the previous subsection. However, as the backbones of our analysis hold even when $N < N^*$, we only invoke $N \geq N^*$ when necessary.

Our key assumption is that players can only use *monotonic strategies*: mappings $[0, 1] \mapsto M$ for the Sender and $M \mapsto [0, 1]$ for the Receiver that are (weakly) increasing. We study the *monotonic cheap-talk game* in which players' (pure) strategy sets are the monotonic strategies. Denote the Receiver's (pure) monotonic strategy set by $\mathcal{A}$ and the Sender's (pure) monotonic strategy set by $\mathcal{S}$.

As discussed in the introduction, we view imposing such monotonicity as capturing a shared language convention. Given the Sender's upward bias, we view it as plausible that higher types must use higher messages, and the Receiver must interpret higher messages accordingly. While the original game allows for arbitrary permutations of messages, we restrict attention to strategies where the ordinal ranking of messages is preserved in use and interpretation.

Monotonicity by itself does not alter the set of equilibrium outcomes. Given any equilibrium of the original game with cutoffs $0 = t_0 < t_1 < \dots < t_n = 1$, the same outcome obtains in an equilibrium of the monotonic cheap-talk game. In particular, we can support the outcome using only the highest messages: for each $i \in \{1, \dots, n\}$, the pool (t_{i-1}, t_i) sends message $m_{N-(n-i)}$, and the Receiver responds to message m_i with $a^R(t_{i-1}, t_i)$. Off path, the Receiver can, for example, respond to any message m_i (where $i < N - n + 1$) with the lowest on-path action, $a^R(0, t_1)$.

[NK: Please make sure you happy with this and next paragraph (in particular, my framing of “less important” and “more fundamental”).] Nevertheless, focusing on monotonic strategies is restrictive in two ways. The less-important aspect is that either player may

have non-monotonic best responses to a monotonic strategy of the opponent. This is due to indifferences. Specifically, the Sender’s (unique) best response to any strictly monotonic Receiver strategy is monotonic, and the Sender always has a monotonic best response to any monotonic Receiver strategy. But if the Receiver takes the same action after two messages, the Sender could optimally choose between them non-monotonically. Similarly, the Receiver has a monotonic best response to any monotonic Sender strategy, and any Receiver best response must be monotonic over the messages that the Sender uses with positive probability (i.e., on path). But if there are unused messages, the Receiver could optimally respond to those in a manner than violates monotonicity.

A more fundamental issue is that the Receiver may have only non-monotonic best responses to mixtures of monotonic Sender strategies. Intuitively, this is because the Sender’s message may reveal information about which strategy was played, and monotonicity is only a “within-strategy” property, not one “across strategies”.⁹ This issue is why dominance-based arguments that gain traction in the monotonic cheap-talk game (as we will see) are blunted in the original game.

3 Key Concepts

For our analysis, it is convenient to view monotonic (pure) strategies as N -dimensional vectors, where N is the number of messages. Abusing notation a little, a monotonic Sender strategy is vector of cutoffs $t := (t_0, t_1, \dots, t_N)$ satisfying

$$0 = t_0 \leq t_1 \leq t_2 \leq \dots \leq t_N = 1.$$

Types in the interval $[t_{i-1}, t_i)$ send message m_i , and type 1 sends message m_N . When $t_{i-1} = t_i$, message m_i is *unused* by the strategy t . A monotonic Receiver strategy is a vector of actions $a := (a_1, a_2, \dots, a_N)$ satisfying

$$0 \leq a_1 \leq a_2 \leq \dots \leq a_N \leq 1.$$

Upon receiving message m_i , the Receiver takes action a_i .

Hereafter, “strategy” without qualification should be understood as a “(pure) monotonic

⁹For example, suppose types are uniformly distributed and there are three messages: $m_1 < m_2 < m_3$. Consider two monotonic Sender strategies: one sends m_1 for $t < 0.9$ and m_3 otherwise; the other sends m_1 for $t < 0.95$ and m_2 otherwise. Under a 50-50 mixture of those two strategies, $\mathbb{E}[t \mid m_2] = 0.975 > 0.95 = \mathbb{E}[t \mid m_3]$. Hence, a Receiver with quadratic-loss utility (under which the optimal action is the expected type) only has non-monotonic best responses even on path.

strategy” and viewed as an increasing vector.

3.1 Robust Best Responses

We compare vectors, and hence strategies, using the component-wise order. We use the sup norm on vectors: $\|t - t'\| := \max_i |t_i - t'_i|$ and similarly for actions. Convergence of strategies (e.g., $t \rightarrow t'$ or $a \rightarrow a'$) is with respect to this norm.

Definition 2. A strategy σ for either player is a *robust best response* to opponent strategy ω if for every $\varepsilon > 0$ there exists $\delta > 0$ such that: if $\|\omega' - \omega\| < \delta$, there exists a best response σ' to ω' with $\|\sigma' - \sigma\| < \varepsilon$.

Intuitively, a strategy is a robust best response (RBR) if small perturbations of the opponent’s strategy have nearby best responses. This rules out best responses that are only justified by knife-edged indifferences; in particular, it pins down Receiver actions after unused messages and pins down Sender tie-breaking when there are multiple messages that lead to the same action.

For example, there are many best responses to the Sender’s strategy $(0, 1/2, 1/2, 1)$ because this strategy does not use message m_2 . However, the unique RBR prescribes action $a^R(1/2)$ in response to m_2 , because any best response to a strategy $(0, t_1, t_2, 1) \approx (0, 1/2, 1/2, 1)$ with $t_1 < t_2$ must prescribe an action close to $a^R(1/2)$ after message m_2 . Similarly, any Sender strategy is a best response to the Receiver strategy $(1/2, 1/2, 1/2)$. However, assuming $a^S(0) < 1/2$, which implies $a^S(t_1) = 1/2$ for some $t_1 \in (0, 1)$, the unique Sender RBR is $(0, t_1, t_1, 1)$, i.e., message m_1 for types $[0, t_1)$, and message m_3 for types $[t_1, 1]$.

More generally, the Receiver RBR to Sender strategy $(0 = t_0, \dots, t_N = 1)$ is

$$(a^R(t_0, t_1), \dots, a^R(t_{i-1}, t_i), \dots, a^R(t_{N-1}, t_N)) ,$$

where $a^R(t_{i-1}, t_i)$ is defined in [Equation 1](#) if $t_{i-1} < t_i$ and $a^R(t_{i-1}, t_i) := a^R(t_i)$ if $t_{i-1} = t_i$.

To describe the Sender’s RBR in general, fix a Receiver strategy $a = (a_1, \dots, a_N)$. Each Sender RBR cutoff is the type that is indifferent between adjacent actions, or the relevant boundary type if there is no indifferent type. Formally, for each $i \in \{1, \dots, N - 1\}$, define $\tau(a_i, a_{i+1})$ as follows:

- If $a_i < a_{i+1}$: the unique type indifferent between a_i and a_{i+1} , or 0 if all types prefer a_{i+1} , or 1 if all types prefer a_i .

- If $a_i = a_{i+1}$: the type $(a^S)^{-1}(a_i)$ for whom a_i is the ideal action, or 0 if $a_i < a^S(0)$, or 1 if $a_i > a^S(1)$.

The Sender's RBR to a is $(0, \tau(a_1, a_2), \dots, \tau(a_{N-1}, a_N), 1)$.

Lemma 1. *Each player has a unique robust best response to any opponent strategy, and this robust best response is continuous and increasing in the opponent's strategy.*

Below, we will consider iteration of RBRs. We interpret iteration of any best responses as adaptive learning in a population, not repeated play between a fixed sender and receiver. We imagine that in each period, players observe many interactions, form expectations about opponents' strategies in the next period, and best respond. RBRs provide protection against small mistakes in this process, which are especially plausible when beliefs are formed from noisy population data.

3.2 Bounding Sequences

Our analysis revolves around two sequences of strategy profiles, generated by iterating RBRs from extreme initial conditions. The highest (in vector order) initial conditions are:

$$\bar{t}^0 := (0, 1, 1, \dots, 1) \quad \text{and} \quad \bar{a}^0 := (1, 1, \dots, 1).$$

Under $\bar{t}^0$, all types send message m_1 ; under $\bar{a}^0$, all messages lead to action 1. The lowest initial conditions are:

$$\underline{t}^0 := (0, 0, \dots, 0, 1) \quad \text{and} \quad \underline{a}^0 := (0, 0, \dots, 0).$$

Under $\underline{t}^0$, all types send message m_N ; under $\underline{a}^0$, all messages lead to action 0.

Now iteratively define $(\bar{t}^{k+1}, \bar{a}^{k+1})$ and $(\underline{t}^{k+1}, \underline{a}^{k+1})$ as follows, for integers $k \geq 0$:

$$\begin{aligned} \bar{a}^{k+1} &:= \text{Receiver's RBR to } \bar{t}^k & \text{and} & & \underline{a}^{k+1} &:= \text{Receiver's RBR to } \underline{t}^k, \\ \bar{t}^{k+1} &:= \text{Sender's RBR to } \bar{a}^k & \text{and} & & \underline{t}^{k+1} &:= \text{Sender's RBR to } \underline{a}^k. \end{aligned}$$

Lemma 2. *The sequences $(\bar{t}^k, \bar{a}^k)_{k=0}^\infty$ and $(\underline{t}^k, \underline{a}^k)_{k=0}^\infty$ are ordered, monotonic, and converge to equilibria:*

1. $\bar{t}_i^k \geq \underline{t}_i^k$ and $\bar{a}_i^k \geq \underline{a}_i^k$ for all i, k ;
2. $\bar{t}_i^k \geq \bar{t}_i^{k+1}$ and $\bar{a}_i^k \geq \bar{a}_i^{k+1}$ for all i, k ;
3. $\underline{t}_i^k \leq \underline{t}_i^{k+1}$ and $\underline{a}_i^k \leq \underline{a}_i^{k+1}$ for all i, k ;

4. $\bar{t}^k \rightarrow \bar{t}^*$, $\bar{a}^k \rightarrow \bar{a}^*$, $\underline{t}^k \rightarrow \underline{t}^*$, and $\underline{a}^k \rightarrow \underline{a}^*$; ¹⁰

5. The limits $(\bar{t}^*, \bar{a}^*)$ and $(\underline{t}^*, \underline{a}^*)$ are equilibria in robust best responses.

These two sequences $(\bar{t}^k, \bar{a}^k)_{k=0}^\infty$ and $(\underline{t}^k, \underline{a}^k)_{k=0}^\infty$ underpin both of our selection arguments. In particular, for our learning-based selection, we will show that any robust best-response sequence starting from arbitrary initial conditions is sandwiched between these bounds; hence when the bounds have a common limit, so does every RBR sequence.

3.3 Dominance

We define the dominance notion here; the formal iterated deletion procedure and results are in [Section 5.2](#).

Definition 3. Fix sets of monotonic strategies $\mathcal{S}' \subseteq \mathcal{S}$ and $\mathcal{A}' \subseteq \mathcal{A}$.

1. A Receiver strategy $a \in \mathcal{A}'$ *interim (weakly) dominates* $a' \in \mathcal{A}'$ relative to $\mathcal{S}'$ if:
 - (a) for every $t \in \mathcal{S}'$ and every message m_i sent with positive probability under t (i.e., $t_{i-1} < t_i$), action a_i yields higher expected payoff than a'_i conditional on the pool of types sending m_i ; and
 - (b) for at least one such pair (t, m_i) , the expected payoff is strictly higher from a_i than a'_i .
2. A Sender strategy $t \in \mathcal{S}'$ *interim (weakly) dominates* $t' \in \mathcal{S}'$ relative to $\mathcal{A}'$ if:
 - (a) for every $a \in \mathcal{A}'$ and almost every type $\tau \in [0, 1]$, type τ prefers the induced action under t to that under t' ; and
 - (b) there exist $a \in \mathcal{A}'$ and a positive-measure set $E \subseteq [0, 1]$ such that, against a , every type $\tau \in E$ strictly prefers the induced action under t to that under t' .

It is natural to consider interim (rather than ex-ante) dominance because the Sender observes his type while the Receiver observes the message. [NK: *Is there a connection to make with Shimoji and Watson's "conditional dominance"?*] For both players, interim dominance implies usual weak dominance.¹¹ The converse is not generally true because interim dominance does not allow for the compensation across messages/types that usual dominance

¹⁰ In other words, $\bar{t}^k$ converges, with its limit denoted $\bar{t}^*$, and so on.

¹¹ Let $U^i(t, a)$ be player i 's expected payoff from the strategy profile $(t, a) \in \mathcal{S} \times \mathcal{A}$. Fix any $\mathcal{S}' \subseteq \mathcal{S}$ and $\mathcal{A}' \subseteq \mathcal{A}$. A Receiver strategy $a \in \mathcal{A}'$ (weakly) dominates a strategy $a' \in \mathcal{A}'$ if $U^R(t, a) \geq U^R(t, a')$ for every $t \in \mathcal{S}'$, with strict inequality for some $t \in \mathcal{S}'$. Sender dominance is analogous.

permits. Consequently, in general, the strategies that survive a process of IDIWDS may still be further refined by usual weak dominance. However, if there is a unique surviving strategy profile under IDWIDS—which will be our focus—then that process also yields the same selection under iterated deletion of weakly dominated strategies (IDWDS).

We will see that the limits $(\bar{t}^*, \bar{a}^*)$ and $(\underline{t}^*, \underline{a}^*)$ characterize the set of strategies that survive a process of IDIWDS. Indeed, we will roughly show that at each round of deletion $k \geq 0$, strategies outside the bounds $(\bar{t}^k, \bar{a}^k)$ and $(\underline{t}^k, \underline{a}^k)$ are interim dominated at that round. The formal treatment in [Section 5.2](#) is more involved because it must overcome an issue that the Sender’s robust best responses may leave low messages unused, while dominance arguments for the Receiver require on-path messages as witnesses.

4 A Two-Message Uniform-Quadratic Example

Before presenting the formal results, we illustrate the main ideas using the uniform-quadratic specification with bias $b \in (1/12, 1/4)$. This is the range in which the underlying CS cheap-talk game has precisely two equilibrium outcomes: an uninformative one in which action 0.5 is induced by all types; and an informative outcome in which types below $1/2 - 2b$ induce action $1/4 - b$, while types above $1/2 - 2b$ induce action $3/4 - b$.

Consider the monotonic cheap-talk game with two messages: $M = \{m_1, m_2\}$, with $m_1 < m_2$. A Sender strategy is then described by a single cutoff $t_1 \in [0, 1]$: types below t_1 send m_1 , and types above send m_2 . A Receiver strategy is $a = (a_1, a_2) \in [0, 1]^2$ with $a_1 \leq a_2$, where each a_j is the response to message m_j .

4.1 Robust Best-Response Iteration

For the uniform-quadratic specification, [Section 3](#)’s robust best-response iteration from any (a^k, t^k) yields:

$$a_1^{k+1} = \frac{t_1^k}{2}, \quad a_2^{k+1} = \frac{t_1^k + 1}{2}, \quad (4)$$

$$t_1^{k+1} = \max \left\{ 0, \frac{a_1^k + a_2^k}{2} - b \right\}. \quad (5)$$

The equations in (4) reflect that the Receiver’s optimal action after each used message is her conditional expectation of the Sender’s type. [Equation 5](#) reflects that the indifferent type (if it exists) is b below the midpoint of the actions, with the max operator accounting for the

possibility that all types prefer the higher action.¹²

Substituting Equation 4 into Equation 5 and vice versa (and using $b < 1/4$) yields the composite maps:

$$t_1^{k+2} = \frac{t_1^k}{2} + \frac{1}{4} - b, \quad (6)$$

$$a_1^{k+2} = \frac{a_1^k}{2} + \frac{1}{8} - \frac{b}{2}, \quad a_2^{k+2} = \frac{a_2^k}{2} + \frac{3}{8} - \frac{b}{2}. \quad (7)$$

These composite maps are contractions with slope $1/2$. Their unique fixed points are

$$t_1^* = \frac{1}{2} - 2b, \quad a_1^* = \frac{1}{4} - b, \quad a_2^* = \frac{3}{4} - b,$$

which match the informative equilibrium outcome. This means that an uninformative equilibrium is not the limit of any robust best-response sequence, no matter the initial conditions. In particular, although the uninformative Sender strategy with $t_1 \in \{0, 1\}$ is a best response to the constant Receiver strategy $a = (0.5, 0.5)$, the Sender's (unique) robust best response is the informative strategy $t_1 = 1/2 - b$, to which $a = (0.5, 0.5)$ is not a best response.

We next consider our two bounding sequences.¹³ The lower bounding sequence $(\underline{t}^k, \underline{a}^k)$ starts from $\underline{t}_1^0 = 0$ and $\underline{a}^0 = (0, 0)$; the upper bounding sequence $(\bar{t}^k, \bar{a}^k)$ starts from $\bar{t}_1^0 = 1$ and $\bar{a}^0 = (1, 1)$. Since $\underline{t}_1^0 = 0 < t_1^*$, $\underline{a}_1^0 = 0 < a_1^*$, and $\underline{a}_2^0 = 0 < a_2^*$, the lower sequences increase towards their fixed points; since $\bar{t}_1^0 = 1 > t_1^*$, $\bar{a}_1^0 = 1 > a_1^*$, and $\bar{a}_2^0 = 1 > a_2^*$, the upper sequences decrease toward them. We see from (6) and (7) that any robust best-response sequence is sandwiched between these two sequences. Table 1 illustrates the two sequences numerically when $b = 0.1$.

4.2 Iterated Deletion of Dominated Strategies

We now explain how the bounding sequences also characterize a process of iterated deletion of interim (weakly) dominated strategies.

Round 1: We assess interim dominance relative to all monotonic strategies, $\mathcal{S}$ and $\mathcal{A}$.

¹²Note that the restriction to monotonic strategies is important for robust best responses to be well-defined. Without that, there would be no robust best response to the Sender strategy $t_1 = 0$ (all types send m_2): perturbing to a strategy where types near 0 send m_1 would require $a_1 \approx 0$, while perturbing so that types near 1 send m_1 would require $a_1 \approx 1$.

¹³In this example, we already established that all robust best response sequences have a common limit. What is more general is that the bounding sequences sandwich all robust best responses sequences, even away from the limit. Moreover, the bounding sequences are crucial for our iterated dominance arguments, even in this example.

k	$\underline{t}_1^k$	$\underline{a}^k$	$\bar{t}_1^k$	$\bar{a}^k$
0	0	(0, 0)	1	(1, 1)
1	0	(0, 0.5)	0.9	(0.5, 1)
2	0.15	(0, 0.5)	0.65	(0.45, 0.95)
4	0.225	(0.075, 0.575)	0.475	(0.3, 0.8)
6	0.2625	(0.1125, 0.6125)	0.3875	(0.2375, 0.7375)
limit	0.3	(0.15, 0.65)	0.3	(0.15, 0.65)

Table 1: Bounding sequences for $b = 0.1$, $N = 2$. For brevity, we display only some iterates.

- For the Receiver, any strategy with $a_2 < \underline{a}_2^1 = 0.5$ is interim dominated: against any Sender strategy that uses m_2 (i.e., when $t_1 < 1$), action 0.5 is strictly better than any $a_2 < 0.5$ after message m_2 , because the message reveals the Sender's type is in $[t_1, 1]$. Symmetrically, any strategy with $a_1 > \bar{a}_1^1 = 0.5$ is interim dominated. So we can delete all Receiver strategies except those with $a_1 \leq 0.5 \leq a_2$, or equivalently, $\underline{a}^1 \leq a \leq \bar{a}^1$.¹⁴
- For the Sender, all types strictly above $\bar{t}_1^1 = 1 - b$ weakly prefer the higher action for all Receiver strategies, and strictly so whenever $a_1 < a_2$. Any cutoff $t_1 > \bar{t}_1^1$ is thus interim dominated by $\bar{t}_1^1$, noting that types below $\bar{t}_1^1$ send the same message under both cutoffs t_1 and $\bar{t}_1^1$. So we can delete all Sender strategies except those with $0 = \underline{t}_1^1 \leq t_1 \leq \bar{t}_1^1$.¹⁵

Round 2: We assess interim dominance relative to the strategies surviving from round 1. For the Receiver, given $t_1 \leq \bar{t}_1^1 = 1 - b$, the mean of types sending m_1 is at most $(1 - b)/2$ and the mean of types sending m_2 is at most $1 - b/2$, so any strategy with $a_1 > (1 - b)/2 = \bar{a}_1^2$ or $a_2 > 1 - b/2 = \bar{a}_2^2$ is interim dominated. Hence, we delete all Receiver strategies except those in $[\underline{a}^2, \bar{a}^2]$. For the Sender, given $a_2 \geq 1/2 = \underline{a}_2^1$, any $t_1 < 1/4 - b = \underline{t}_1^2$ is interim dominated; given $a_1 \leq 1/2 = \bar{a}_1^2$, any $t_1 > 3/4 - b = \bar{t}_1^2$ is interim dominated. Hence we delete all Sender strategies except those with in $[\underline{t}_1^2, \bar{t}_1^2]$.

Subsequent rounds: Reasoning analogously, in each round $k > 1$, interim domination relative to the surviving strategies from the previous round deletes Receiver strategies outside $[\underline{a}^k, \bar{a}^k]$ and Sender strategies outside $[\underline{t}_1^k, \bar{t}_1^k]$. The limit of this deletion process yields the unique survivor (t_1^*, a^*) .

¹⁴No strategy within these bounds is interim dominated at this stage. For any distinct $a_1, a'_1 \in [0, 0.5]$, there is a Sender strategy against which a_1 is strictly better than a'_1 after m_1 , and vice-versa. An analogous point applies to $a_2, a'_2 \in [0.5, 1]$. Note also that absent the restriction to monotonic Sender strategies, no Receiver strategy would be interim dominated.

¹⁵No Sender strategy within these bounds is interim dominated at this stage. Raising the cutoff from $t_1 \in [0, 1 - b]$ to $t'_1 > t_1$ hurts types in (t_1, t'_1) whenever $a_1 < a_2$ and $(a_1 + a_2)/2 < t'_1 + b$; lowering the cutoff to $t'_1 < t_1$ hurts types in (t'_1, t_1) whenever $a_1 < a_2$ and $(a_1 + a_2)/2 > t'_1 + b$.

Although this is a specific procedure of IDIWDS, it is a natural one: in each round, we delete all the interim dominated strategies for each player.¹⁶ In this example, we can show that the order of deletion does not matter; more generally, we do not have such a proof. Furthermore, when we allow for additional messages—in particular, when $N > N^*$ —there are nuances that our general analysis in [Section 5.2](#) handles.

5 Results

5.1 Robust Best-Response Iteration

We write $\liminf$ and $\limsup$ of vectors in the component-wise sense. Recall that a sequence of robust best responses $(t^k, a^k)_{k \geq 0}$ is pinned down by its initial conditions (t^0, a^0) .

Theorem 1. *For any robust best-response sequence $(t^k, a^k)_{k=0}^\infty$:*

$$\underline{t}^k \leq t^k \leq \bar{t}^k \quad \text{and} \quad \underline{a}^k \leq a^k \leq \bar{a}^k \quad \text{for all } k,$$

and consequently

$$\underline{t}^* \leq \liminf_{k \rightarrow \infty} t^k \leq \limsup_{k \rightarrow \infty} t^k \leq \bar{t}^* \quad \text{and} \quad \underline{a}^* \leq \liminf_{k \rightarrow \infty} a^k \leq \limsup_{k \rightarrow \infty} a^k \leq \bar{a}^*.$$

The theorem says that starting from arbitrary initial conditions, the sequence generated by robust best-response iteration is sandwiched between $(\underline{t}^k, \underline{a}^k)_{k \geq 0}$ and $(\bar{t}^k, \bar{a}^k)_{k \geq 0}$, and hence the players' strategies only accumulate within $[\underline{t}^*, \bar{t}^*]$ and $[\underline{a}^*, \bar{a}^*]$. That is, play is asymptotically bounded by the two equilibria $(\underline{t}^*, \underline{a}^*)$ and $(\bar{t}^*, \bar{a}^*)$.¹⁷

While in general the two bounding equilibria can be distinct, even in terms of outcomes, they coincide under a standard condition. To establish that, we use the following two properties of the bounding equilibria (indeed, of any equilibrium in which the Receiver's strategy is a robust best response). The first says that all unused messages are at the bottom and elicit action $a^R(0) = 0$, and the second says that they satisfy NITS when $N \geq N^*$.

¹⁶ The “interim” qualification is important. As noted in [Section 3.3](#), all interim dominated strategies are (weakly) dominated. So the current deletion process also yields selection via iterated deletion of dominated strategies. But there may be dominated strategies that are not interim dominated at any given round (hence not deleted in that round by our process), because of “cross-compensation” across types for the Sender or across messages for the Receiver. For example, one can check that in the first round, any Receiver strategy with $a_2 > a_1 + 1/2$ is dominated; but it is not interim dominated as explained in [footnote 14](#).

¹⁷ Although it is not implied by the theorem, one can show that any robust best-response sequence converges even when $(\underline{t}^*, \underline{a}^*) \neq (\bar{t}^*, \bar{a}^*)$; see [Olszewski \(2022\)](#).

Lemma 3. *Let (t^*, a^*) be an equilibrium in which a^* is a robust best response. The set of messages unused by t^* is a lower set, and a^* responds to all those messages with action $a^R(0) = 0$.*

The idea is that if t^* has an unused message m_i above a used message m_k , then a^* being a robust best response to t^* pins down its off-path action at $a_i^* = a^R(t_i^*) > a_k^*$. This in turn implies that the off-path action a_i^* is either strictly between two consecutive on-path actions, or strictly above all on-path actions. Either way, some types would deviate to message m_i , a contradiction.

Lemma 4. *Let (t^*, a^*) be an equilibrium in which a^* is a robust best response. If $N \geq N^*$, then (t^*, a^*) satisfies NITS.*

This lemma follows from the previous one if t^* has unused messages. If t^* uses all messages, then (t^*, a^*) induces N^* actions, and any such equilibrium satisfies NITS.

[NK: First part of next result (which I have now labeled a proposition) is new.]

Proposition 1. *Assume $N \geq N^*$.*

1. *Any robust best-response sequence converges to an equilibrium that satisfies NITS, and conversely any NITS equilibrium outcome is the outcome of the limit of some robust-best response sequence.*
2. *If there is a unique equilibrium outcome satisfying NITS, then all robust best-response sequences converge to the same equilibrium, namely $(t^*, a^*) := (\underline{t}^*, \underline{a}^*) = (\bar{t}^*, \bar{a}^*)$, which uses the N^* highest messages.*

Proof of Proposition 1. (Part 1.) Olszewski (2022, Corollary 1) implies that any robust best-response sequence converges; the limit is an equilibrium in robust best responses by the continuity of robust best responses (Lemma 1). It follows from Lemma 4 that the limit equilibrium satisfies NITS. Conversely, take any NITS equilibrium outcome with n induced actions. Specify the initial conditions such that (t^0, a^0) induces that outcome with the n highest messages used, and $a_i^0 = 0$ for any unused message m_i (i.e., $i \leq N - n$). Robust best response iteration from these initial conditions is constant.

(Part 2.) Now assume a unique equilibrium outcome satisfying NITS. Let $(t^*, a^*) \in \{(\underline{t}^*, \underline{a}^*), (\bar{t}^*, \bar{a}^*)\}$. Since (t^*, a^*) satisfies NITS (by the first part), Chen, Kartik and Sobel (2008, Proposition 1) implies that (t^*, a^*) induces N^* actions (otherwise, there would be multiple equilibrium outcomes satisfying NITS). Since all messages used by t^* induce distinct

actions under a^* , it follows from [Lemma 3](#) that t^* uses the N^* highest messages and $t_i^* = a_i^* = 0$ for $i \in \{1, \dots, N - N^*\}$. Since (t^*, a^*) satisfies NITS and could have been either $(\underline{t}^*, \underline{a}^*)$ or $(\bar{t}^*, \bar{a}^*)$, the hypothesis of a unique NITS outcome implies $(\underline{t}^*, \underline{a}^*) = (\bar{t}^*, \bar{a}^*)$. The result now follows from [Theorem 1](#). $\square$

The first part of the proposition says that, given enough messages, robust best response iteration from arbitrary initial conditions selects only the NITS equilibrium outcomes, but all of them. This always includes all the equilibrium outcomes with N^* actions, and under some conditions, rules all outcomes with fewer actions than some threshold ([Chen, Kartik and Sobel, 2008](#), Propositions 1 and 2). The second part of the proposition further says if there is a unique NITS equilibrium outcome—e.g., when CS’ regularity condition holds—then robust best response iteration selects not just the unique N^* equilibrium outcome but also a unique equilibrium. In this equilibrium, the Sender uses maximal exaggeration (i.e., the N^* highest messages) to induce the N^* actions.

[NK: *Next remark is new. It could be stated in other ways too.*]

Remark 1. [Theorem 1](#), and hence [Proposition 1](#), does not hinge on assuming from the outset that players use monotonic strategies. Take any initial Receiver action profile $a^0 = (a_1^0, \dots, a_N^0)$, which need not be monotonic with respect to any exogenous order. We may (re-)define the order on messages such that $a_1^0 \leq \dots \leq a_N^0$. We can then iterate robust best responses. The Sender’s robust best response to any a^k is monotonic: if $a_i^k > a_j^k$, the set of types who prefer a_i^k to a_j^k is an upper interval, and the robust best response tie-breaking preserves monotonicity across blocks of identical actions. In turn, the Receiver’s robust best response to a monotonic Sender strategy is monotonic. Therefore, the sandwiching and limit arguments of [Theorem 1](#) apply, as does [Proposition 1](#), with respect to the order defined by a^0 . Thus, when there is a unique NITS outcome (and when $N \geq N^*$), robust best response iteration from arbitrary initial conditions always converges to an equilibrium with that outcome; the messages used will be the N^* highest messages according to a^0 .

5.2 Iterated Deletion of Dominated Strategies

We present a result parallel to [Theorem 1](#) for iterated deletion of interim dominated strategies. That is, we offer a process of iterative deletion of interim dominated strategies that leads to the strategy sets

$$\mathcal{A}^* := \{\underline{a}^* \leq a \leq \bar{a}^*\} \quad \text{and} \quad \mathcal{S}^* := \{\underline{t}^* \leq a \leq \bar{t}^*\}.$$

The idea of the deletion process is intuitive: roughly speaking, in every round $k \geq 0$ we delete the strategies outside $[\underline{a}^k, \bar{a}^k]$ and $[\underline{t}^k, \bar{t}^k]$. The nuance is that at some stage the upper bound $\bar{t}^k$ may have unused low messages (i.e., $\bar{t}_i^k = 0$ for some i), in which case those messages are unused by all strategies in $[\underline{t}^k, \bar{t}^k]$, and interim dominance then has no bite at those unused messages. Our argument thus involves perturbing the upper bounds. (These perturbations are not needed when $N \leq N^*$, as in the example of [Section 4](#).)

Concretely, initialize $\mathcal{A}^0 := \mathcal{A}$ and $\mathcal{S}^0 := \mathcal{S}$ as the sets of all monotonic Receiver and Sender strategies. Let $(\varepsilon_k)_{k=0}^\infty$ be a sequence such that each $\varepsilon_k \in (0, 1)$ and $\varepsilon_k \rightarrow 0$. We formalize in [Appendix C.1](#) a construction of sequences $(\tilde{t}^k)_{k \geq 0}$ and $(\tilde{a}^k)_{k \geq 0}$ that are perturbations of $(\bar{t}^k)_{k \geq 0}$ and $(\bar{a}^k)_{k \geq 0}$. Each $\tilde{t}^k$ is a strictly increasing vector (so all messages are used) with $\|\tilde{t}^k - \bar{t}^k\| \leq \varepsilon_k$, and $\tilde{a}^k$ is the robust best response to $\tilde{t}^k$. The iterative deletion process is that for each $k \geq 0$:

(Rk) $\mathcal{A}^{k+1}$ is obtained from $\mathcal{A}^k$ by deleting every a such that $a \notin [\underline{a}^{k+1}, \tilde{a}^{k+1}]$.

(Sk) $\mathcal{S}^{k+1}$ is obtained from $\mathcal{S}^k$ by deleting every t such that $t \notin [\underline{t}^{k+1}, \tilde{t}^{k+1}]$

We refer to this process of iterated deletion as the *bounding-sequence deletion procedure*. Define its limit survivor sets

$$\mathcal{A}^\infty := \bigcap_{k \geq 0} \mathcal{A}^k \quad \text{and} \quad \mathcal{S}^\infty := \bigcap_{k \geq 0} \mathcal{S}^k.$$

Theorem 2. *Consider the bounding-sequence deletion procedure.*

1. *For all $k \geq 0$: each Receiver strategy in $\mathcal{A}^k \setminus \mathcal{A}^{k+1}$ is interim dominated by some strategy in $\mathcal{A}^k$ relative to $\mathcal{S}^k$; and analogously for the Sender.*
2. *The limit survivor sets are $\mathcal{A}^\infty = \mathcal{A}^*$ and $\mathcal{S}^\infty = \mathcal{S}^*$.*
3. *No Receiver strategy in $\mathcal{A}^*$ is interim weakly dominated by another strategy in $\mathcal{A}^*$ relative to $\mathcal{S}^*$, and analogously for the Sender.*

This theorem says that the bounding-sequence deletion procedure is a *valid* process of IDIWDS in the sense that it deletes only interim dominated strategies at each stage, the process continues so long as there are any interim dominated strategies, and the limit sets are nonempty. Since interim dominance implies dominance, the process also only deletes dominated strategies at each stage. However, it is possible in general that $(\mathcal{S}^\infty, \mathcal{A}^\infty)$ contains strategies that are dominated but not interim dominated. That is not an issue if $(\mathcal{S}^\infty, \mathcal{A}^\infty)$ is a singleton; in that case our procedure is also a valid process of IDWDS.

Proposition 2. *Assume $N \geq N^*$ and a unique equilibrium outcome satisfying NITS. The bounding-sequence deletion procedure is a valid process of IDIWDS (and also IDWDS) that yields a unique strategy profile. That strategy profile is the same as in [Proposition 1](#) part 2, which is an equilibrium that satisfies NITS and induces N^* actions.*

Proof. [Proposition 1](#) established that, under the current hypotheses, $(\mathcal{S}^*, \mathcal{A}^*) = \{(\bar{t}^*, \bar{a}^*)\} = \{(\underline{t}^*, \underline{a}^*)\}$, and this profile is an equilibrium that satisfies NITS and thus induces N^* actions. The result follows from [Theorem 2](#). $\square$

We conjecture that results similar to [Theorem 2](#) and [Proposition 2](#) can be obtained for any valid ID(I)WDS procedure, not just the bounding-sequence deletion procedure. The difficulty in establishing that stems from the multiplicity of best responses.

[NK: *I dropped the 3-message UQ example entirely. In my view, it is too tedious and does not add enough now that we have the two-message UQ example earlier (which I think is great—thanks Wojciech) and also touch elsewhere on some of the points the 3-message example made (which I think were anyway likely to be lost in the 3-message example).*]

6 Related Literature

[NK: *lots of changes (and abbreviation) to this section; please make sure you are happy with it. Melody: please confirm if not having “Melody” in the bib file’s author fields for your sole-authored papers is correct.*]

This paper unifies ideas from earlier working papers by different subsets of the authors: [Lo \(2007\)](#), [Gordon \(2010, 2011\)](#), [Kartik and Sobel \(2015\)](#), and [Lo and Olszewski \(2018\)](#). Our goal here is to present transparently the most significant conclusions from those working papers. Those working papers contain analyses for other structures of conflict of interest, and other results that are more general, or simply different, in certain directions.

Iterating best responses and (weak or strict) dominance are classical ideas. Our contribution is to combine these ideas, suitably adapted for cheap talk, with the restriction to monotonic strategies and deduce the implications in the CS ([Crawford and Sobel, 1982](#)) model. The closest published papers are [Sobel \(2019\)](#) and [Olszewski \(2022\)](#), which both stemmed from the work described here. [Olszewski \(2022\)](#) shows, as a consequence of his more general results on sequences of iterations, that best-response iteration converges in our setting—a fact we use in [Proposition 1](#)—but does not study properties of the limit. [Sobel \(2019\)](#) extends [Milgrom and Roberts’ \(1990\)](#) arguments for supermodular games and iterated

deletion of strictly dominated strategies to a broader class of games and iterated deletion of weakly dominated strategies (IDWDS). He points out that a class of monotonic cheap-talk games¹⁸ satisfy a weak form of supermodularity that makes it possible to bound the set of strategies that survive a process of IDWDS. His arguments are related to ours, although his deletion process is phrased via smallest and largest best responses, rather than robust best responses.¹⁹ Importantly, he does not characterize the bounds.

Sobel (2017) and Lo (2021)—the latter stemming from Lo (2009)—apply iterated dominance to pre-play communication in complete-information games, with strategy restrictions on the use of language.

Blume (2025) introduces a notion of language equilibrium in cheap-talk games by iterating best replies starting from a distinguished receiver strategy, interpreted as a pre-existing language. Similar to our approach, language equilibria resolve message indeterminacy and feature language inflation in finite versions of the CS setting. Our approach differs in that selection under robust best responses arises from arbitrary initial conditions; we also discuss iterated dominance.

Farrell (1993) was the first to incorporate exogenous meaning of language into cheap-talk games. His notion of neologism-proof equilibrium is based on the idea that messages have commonly-accepted meanings that are followed so long as they are consistent with incentives. Neologism-proofness does refine the set of equilibria, but it lacks general existence properties—in particular, even in the uniform-quadratic specification of CS.

The literature contains various selection arguments for the equilibrium outcome with N^* actions in the CS model, which we now turn to.

Mensch (2020) studies existence of monotone pure-strategy perfect Bayesian equilibrium in dynamic games. His main result concerns existence of equilibria with certain monotonicity of beliefs even off path, roughly a form of “support monotonicity”. He notes that such off-path monotonicity selects NITS equilibria in the CS model. Our monotonic-strategies restriction is weaker—hence it has no selection power over outcomes by itself—but robust best responses deliver similar off-path implications. Mensch (2020) does not study iteration of robust best responses or dominance.

Kartik (2009) and Chen (2011) assume the message space equals the type space and introduce perturbations motivated by exogenous meaning: lying costs in Kartik’s case, and

¹⁸He assumes the prior is finitely supported, which technically does not subsume our setting.

¹⁹This distinction is not crucial; our arguments could also be made via smallest and largest best responses.

behavioral types in Chen's. When perturbations vanish in both these approaches, equilibria in monotonic strategies converge to NITS outcomes with inflated or exaggerated language, as shown in [Chen, Kartik and Sobel \(2008\)](#). Our approach in this paper achieves a similar selection, but we impose monotonicity on strategies directly and then apply learning/dominance arguments rather than payoff perturbations.

[Dilmé \(2022\)](#) selects CS equilibrium outcomes that are robust to a class of message-cost payoff perturbations. He shows that under the CS regularity condition, only the outcome with the maximal number of induced actions is robust in this sense.²⁰ Dilmé's argument, like ours and those of the papers in the last two paragraphs, operates by showing that some unused messages must lead to low off-path actions. Dilmé does not resolve message indeterminacy.

[Sémirat and Forges \(2023\)](#) study finite cheap-talk games and show that a dynamic adjustment process converges to an undefeated equilibrium in the sense of [Mailath, Okuno-Fujiwara and Postlewaite \(1993\)](#). Similar to us, their process converges to the largest equilibrium, but the approaches differ: their finite type space allows a fully revealing strategy, which they use as their initial condition, and their adjustments are better replies rather than best replies. They do not discuss dominance arguments.

A Proofs for [Section 3](#)

Proof of [Lemma 1](#). First consider the Receiver. For any $t_{i-1} \leq t_i$, the optimal action $a^R(t_{i-1}, t_i)$ is uniquely defined by strict concavity of $u^R(\cdot, \tau)$ for each τ , and it is increasing in both t_{i-1} and t_i by the supermodularity of u^R . By the theorem of the maximum, it is also continuous in (t_{i-1}, t_i) . Hence for any $t = (t_0, t_1, \dots, t_N)$, the strategy

$$(a^R(t_0, t_1), \dots, a^R(t_{i-1}, t_i), \dots, a^R(t_{N-1}, t_N))$$

is a RBR, which is continuous and increasing in t . It is in fact the unique RBR, because for any t' that is strictly increasing, there is a unique best response to t' ; and any RBR must coincide with the limit of those best responses as $t' \rightarrow t$, which is the foregoing RBR.

Now consider the Sender. For any Receiver strategy $a = (a_1, \dots, a_N)$, the strategy $(0 = t_0, \tau(a_1, a_2), \dots, \tau(a_{N-1}, a_N), t_N = 1)$ is an increasing sequence and a RBR. That it is continuous and increasing in a follows from continuity and monotonicity of $\tau(\cdot, \cdot)$, by the

²⁰He also extends this result when the regularity condition fails or the Sender does not have an upward (or downward) bias.

construction described before the lemma. The strategy is also the unique RBR because for any a' that is strictly increasing, there is a unique best response vector t' ; and any RBR must coincide with the limit of those best responses as $a' \rightarrow a$, which is the foregoing RBR. $\square$

Proof of Lemma 2. Part (1): Lemma 1 shows that each player RBR is monotonic in the opponent's strategy. The result follows from induction, given that at $k = 0$, it holds because $\bar{t}^0$ and $\bar{a}^0$ are the highest strategies while $\underline{t}^0$ and $\underline{a}^0$ are the lowest.

Part (2): Since $\bar{t}^0$ and $\bar{a}^0$ are the highest strategies, clearly $\bar{a}^1 \leq \bar{a}^0$ and $\bar{t}^1 \leq \bar{t}^0$. Thereafter we use induction: if $\bar{t}^k \leq \bar{t}^{k-1}$, then $\bar{a}_i^{k+1} = a^R(\bar{t}_{i-1}^k, \bar{t}_i^k) \leq a^R(\bar{t}_{i-1}^{k-1}, \bar{t}_i^{k-1}) = \bar{a}_i^k$, where the inequality uses the same argument as for monotonicity of the Receiver's RBR. Similarly, following the monotonicity for the Sender's RBR, we get that if $\bar{a}^k \leq \bar{a}^{k-1}$, then $\bar{t}^{k+1} \leq \bar{t}^k$.

Part (3) is analogous to part (2).

Part (4) follows because monotone bounded sequences converge.

Part (5) follows because each player's RBR is continuous in the opponent's strategy (Lemma 1), so limits of RBR sequences are mutual robust best responses. $\square$

B Proofs for Section 5.1

Proof of Theorem 1. Fix any initial conditions (t^0, a^0) and the sequence of RBR iterations $(t^k, a^k)_{k \geq 0}$.

We first prove by induction that for all $k \geq 0$, there is sandwiching:

$$\underline{t}^k \leq t^k \leq \bar{t}^k \quad \text{and} \quad \underline{a}^k \leq a^k \leq \bar{a}^k.$$

The base case is immediate from the definitions of the bounding sequences' extreme initial conditions: $\underline{t}^0 \leq t^0 \leq \bar{t}^0$ and $\underline{a}^0 \leq a^0 \leq \bar{a}^0$. For the inductive step, assume $\underline{t}^k \leq t^k \leq \bar{t}^k$ and $\underline{a}^k \leq a^k \leq \bar{a}^k$. The monotonicity of the receiver's RBR to the sender's strategy (Lemma 1) implies that $\underline{a}^{k+1} \leq a^{k+1} \leq \bar{a}^{k+1}$, since (by construction) each of these is the RBR to sender strategies with the same ordering. Analogously, by monotonicity of the sender's RBR to the receiver's strategy (Lemma 1), we get $\underline{t}^{k+1} \leq t^{k+1} \leq \bar{t}^{k+1}$.

Given the sandwiching for all k , the asymptotic conclusions follow from $\bar{t}^k \rightarrow \bar{t}^*$ and $\underline{t}^k \rightarrow \underline{t}^*$ (which was established in Lemma 2). $\square$

Proof of Lemma 3. Consider any equilibrium (t^*, a^*) in which a^* is a robust best response to

t^* . It suffices to prove the lemma's claim about t^* , as the property of a^* follows from it being a robust best response. So suppose, to contradiction, that the set of unused messages is not a lower set. Then letting m_k be the first used message and m_i the first unused message above that, we have $0 = t_{k-1}^* < t_k \leq t_{i-1}^* = t_i^* \leq 1$. It follows that $a_k^* = a^R(t_{k-1}^*, t_k^*) < a^R(t_i^*) = a_i^*$, where the inequality is by monotonicity of a^R and the equalities are because a^* is a robust best response to t^* . Moreover, the monotonicity of t^* (and a^R) implies that for any used message j , either $a_j^* < a_i^*$ or $a_j^* > a_i^*$. So a_i^* is an off-path action distinct from all on-path actions. If there is a used message above m_i , then a_i^* lies strictly between two consecutive on-path actions, and a positive measure of types near that cutoff (i.e., near the type that is indifferent between those consecutive on-path actions) would deviate to m_i . If there is no used message above m_i , then a_i^* is strictly above all on-path actions, and a positive measure of types near 1 would deviate to m_i . Either way, we contradict (t^*, a^*) being an equilibrium. $\square$

Proof of Lemma 4. Assume $N \geq N^*$ and consider any equilibrium (t^*, a^*) in which a^* is a robust best response. If t^* has unused messages, then the result follows from Lemma 3, since the sender has the option to induce $a^R(0) = 0$. If t^* has no unused messages, then $N = N^*$ and (t^*, a^*) induces N^* distinct actions, so Proposition 1 of Chen, Kartik and Sobel (2008) implies that it satisfies NITS. $\square$

C Proofs for Section 5.2

C.1 Constructing the Perturbed Upper Bounding Sequence

Interim dominance for the Receiver concerns on-path messages (i.e., those sent with positive probability by some relevant Sender strategy). But the upper-bounding robust best response sequence can reach a stage at which it does not use a set of low messages, in which case all the strategies between the upper and lower bound do not use those messages. To ensure that each round admits a surviving Sender strategy that uses every message—so the Receiver's interim comparisons have an on-path witness when needed—we perturb the upper bounding Sender sequence to have strategies that are strictly increasing, which we refer to as “activation”. Let us formalize that.

For any $t = (0 = t_0, \dots, t_N = 1)$, let $j(t) := \min\{i \in \{1, \dots, N\} : t_i > 0\}$ be the first used message. For any $\varepsilon \in (0, 1)$, define a self-map $\text{Act}_\varepsilon : \mathcal{S} \rightarrow \mathcal{S}$ by $\hat{t} \mapsto t$, where t is constructed as follows (with $t_0 = 0$ and $t_N = 1$):

1. $t_i := \hat{t}_i$ for all $i \geq j(\hat{t})$.

2. For $i = j(\hat{t}) - 1, j(\hat{t}) - 2, \dots, 1$ define recursively $t_i := \min\{\varepsilon, t_{i+1}/2\}$.

In other words, Act_ε preserves the cutoffs for all messages used by $\hat{t}$, while perturbing cutoffs for each unused lower message by assigning it a small interval of types.

Lemma 5. *For any $\hat{t}$ and any $\varepsilon \in (0, 1)$, the activated vector $t = \text{Act}_\varepsilon(\hat{t})$ satisfies:*

1. $0 = t_0 < t_1 < \dots < t_N = 1$.
2. $t_i = \hat{t}_i$ for all $i \geq j(\hat{t})$.
3. For all $i < j(\hat{t})$, $t_i \leq \varepsilon$.

Proof. Parts (2) and (3) are immediate. For (1), note that the recursive definition $t_i \leq t_{i+1}/2$ for $i < j(\hat{t})$ implies strict inequalities. $\square$

Lemma 6. *If $0 < \varepsilon' \leq \varepsilon < 1$, then for every $\hat{t} \in \mathcal{S}$, we have $\text{Act}_{\varepsilon'}(\hat{t}) \leq \text{Act}_\varepsilon(\hat{t})$.*

Proof. The two activated vectors coincide from index $j(\hat{t})$ onward. For indices below $j(\hat{t})$, both constructions satisfy the same recursion $t_i = \min\{\varepsilon, t_{i+1}/2\}$ (respectively with ε'). Since $\varepsilon' \leq \varepsilon$, a backward induction yields $\text{Act}_{\varepsilon'}(\hat{t})_i \leq \text{Act}_\varepsilon(\hat{t})_i$ for all $i < j(\hat{t})$. $\square$

Let $\text{RBR}^R(t)$ denote the Receiver robust best response to t , and $\text{RBR}^S(a)$ denote the Sender robust best response to a (as characterized in [Section 3.1](#)).

We define a sequence analogous to the upper bounding sequence from [Section 3.2](#) by iterating robust best responses from the highest initial condition, but with a perturbation through the activation map. The activation level shrinks to zero over rounds, which will allow us to ensure a single valid deletion procedure whose limit equals the unperturbed bound sets.

To that end, fix any sequence $(\varepsilon_k)_{k=1}^\infty$ with each $\varepsilon_k \in (0, 1)$ and $\varepsilon_k \rightarrow 0$ (e.g., $\varepsilon_k := 2^{-k}$). Initialize

$$\tilde{t}^0 := (0, 1, 1, \dots, 1), \quad \tilde{a}^0 := (1, 1, \dots, 1),$$

and for $k \geq 0$ define

$$\tilde{a}^{k+1} := \text{RBR}^R(\tilde{t}^k), \quad \tilde{t}^{k+1} := \text{RBR}^S(\tilde{a}^k), \quad \tilde{t}^{k+1} := \text{Act}_{\varepsilon_{k+1}}(\tilde{t}^{k+1}).$$

We will use the original lower bounding sequences $(\underline{t}^k)_{k \geq 0}$ and $(\underline{a}^k)_{k \geq 0}$ from [Section 3.2](#).

Lemma 7. *For every $k \geq 1$, the strategy $\tilde{t}^k$ is strictly increasing. Consequently, $\tilde{a}^k$ is strictly increasing for every $k \geq 1$.*

Proof. For $k \geq 1$, the strategy $\tilde{t}^k$ is obtained by applying $\text{Act}_{\varepsilon_k}$ to some $\tilde{t}^k$, so it is strictly increasing by [Lemma 5](#). For any strictly increasing t , $\text{RBR}^R(t)$ is strictly increasing because for $0 \leq x < y < z \leq 1$, strict supermodularity of u^R implies $a^R(x, y) < a^R(y, z)$. $\square$

The next lemma parallels [Lemma 2](#).

Lemma 8. *For every $k \geq 0$:*

1. $\underline{t}^k \leq \tilde{t}^k$ and $\underline{a}^k \leq \tilde{a}^k$.
2. $\tilde{t}^{k+1} \leq \tilde{t}^k$ and $\tilde{a}^{k+1} \leq \tilde{a}^k$.
3. $\tilde{t}^k \rightarrow \bar{t}^*$, $\tilde{a}^k \rightarrow \bar{a}^*$.

Consequently, for every coordinate i and all integers $1 \leq r \leq k+1$,

$$\underline{a}_i^r \leq \underline{a}_i^{k+1} \leq \tilde{a}_i^{k+1} \leq \tilde{a}_i^r,$$

and for every cutoff coordinate $i \leq N-1$,

$$\underline{t}_i^r \leq \underline{t}_i^{k+1} \leq \tilde{t}_i^{k+1} \leq \tilde{t}_i^r.$$

Proof. We use that $\text{RBR}^R(\cdot)$, $\text{RBR}^S(\cdot)$, and $\text{Act}_\varepsilon(\cdot)$ are all increasing. We also use [Lemma 6](#) and that (ε_k) is nonincreasing.

Part (1): [Lemma 1](#) shows that each player's robust best response is monotonic, and activation can only increase cutoffs. The result follows from induction, given that $\underline{t}^0 \leq \tilde{t}^0$ and $\underline{a}^0 \leq \tilde{a}^0$ by construction.

Part (2): We use induction. Since $\tilde{t}^0$ and $\tilde{a}^0$ are the highest strategies, we have $\tilde{a}^1 \leq \tilde{a}^0$ and $\tilde{t}^1 \leq \tilde{t}^0$. Assume $\tilde{t}^k \leq \tilde{t}^{k-1}$. Then monotonicity of RBR^R implies $\tilde{a}^{k+1} \leq \tilde{a}^k$, hence by monotonicity of RBR^S ,

$$\tilde{t}^{k+1} = \text{RBR}^S(\tilde{a}^k) \leq \text{RBR}^S(\tilde{a}^{k-1}) = \tilde{t}^k.$$

Applying $\text{Act}_{\varepsilon_{k+1}}$ and using its monotonicity in both arguments gives

$$\tilde{t}^{k+1} = \text{Act}_{\varepsilon_{k+1}}(\tilde{t}^{k+1}) \leq \text{Act}_{\varepsilon_{k+1}}(\tilde{t}^k) \leq \text{Act}_{\varepsilon_k}(\tilde{t}^k) = \tilde{t}^k,$$

where the second inequality uses [Lemma 6](#) and $\varepsilon_{k+1} \leq \varepsilon_k$. This yields part (2).

Part (3): Part (2) implies that $(\tilde{t}^k)_{k \geq 0}$ and $(\tilde{a}^k)_{k \geq 0}$ are bounded monotone sequences, hence they converge. It remains to note that the vanishing activation does not affect the limiting

bounds: the activation only changes a zero-prefix and the magnitude of the change is bounded by $\varepsilon_k \downarrow 0$ (Lemma 5). By continuity of RBR^R and RBR^S (Lemma 1), the limiting relations coincide with those of the unactivated bounding sequences. This yields part (3).

The lemma's final inequalities follow immediately from parts (1) and (2). $\square$

C.2 Elimination Lemmas

Lemma 9. *Fix $k \geq 0$ and $a \in \mathcal{A}^k$. If there exists $i^* \in \{1, \dots, N\}$ such that either $a_{i^*} > \tilde{a}_{i^*}^{k+1}$ or $a_{i^*} < \underline{a}_{i^*}^{k+1}$, then a is interim weakly dominated by some $b \in \mathcal{A}^k$ relative to $\mathcal{S}^k$.*

Proof. We prove the first case ($a_{i^*} > \tilde{a}_{i^*}^{k+1}$); the second is analogous. Let i^* be the smallest index with $a_{i^*} > \tilde{a}_{i^*}^{k+1}$. Define b by $b_{i^*} := \tilde{a}_{i^*}^{k+1}$ and $b_j := a_j$ for $j \neq i^*$. We first show that $b \in \mathcal{A}^k$, and then show that b interim weakly dominates a relative to $\mathcal{S}^k$.

(*Feasibility.*) It suffices to check $b_{i^*-1} \leq b_{i^*} \leq b_{i^*+1}$ (ignoring the irrelevant inequality if $i^* \in \{1, N\}$). By minimality of i^* and monotonicity of both a and $\tilde{a}^{k+1}$,

$$b_{i^*-1} = a_{i^*-1} \leq \tilde{a}_{i^*-1}^{k+1} \leq \tilde{a}_{i^*}^{k+1} = b_{i^*} \leq a_{i^*} \leq a_{i^*+1} = b_{i^*+1}.$$

So b is monotone.

Since $a \in \mathcal{A}^k$, it satisfies all past bounds $a_i \in [\underline{a}_i^r, \tilde{a}_i^r]$ for all $r \leq k$. The strategy b differs from a only at i^* . By Lemma 8, $\underline{a}^r$ is nondecreasing and $\tilde{a}^r$ is nonincreasing in r , so for every $r \leq k$,

$$\underline{a}_{i^*}^r \leq \underline{a}_{i^*}^{k+1} \leq b_{i^*} \leq \tilde{a}_{i^*}^{k+1} \leq \tilde{a}_{i^*}^r.$$

Hence b satisfies all past bounds and belongs to $\mathcal{A}^k$.

(*Interim dominance.*) Fix any $t \in \mathcal{S}^k$. If m_{i^*} is off-path under t , there is no interim constraint at i^* . If m_{i^*} is on-path (i.e. $t_{i^*-1} < t_{i^*}$), the Receiver's expected payoff conditional on m_{i^*} is strictly concave and maximized at $a^R(t_{i^*-1}, t_{i^*})$. Since $t \leq \tilde{t}^k$, monotonicity of $a^R(\cdot, \cdot)$ yields

$$a^R(t_{i^*-1}, t_{i^*}) \leq a^R(\tilde{t}_{i^*-1}^k, \tilde{t}_{i^*}^k) = \tilde{a}_{i^*}^{k+1} = b_{i^*} < a_{i^*}.$$

Thus replacing a_{i^*} by b_{i^*} weakly improves interim payoff after m_{i^*} for every such t .

For strictness, use the witness Sender strategy $t = \tilde{t}^k \in \mathcal{S}^k$. By Lemma 7, $\tilde{t}^k$ is strictly increasing, so m_{i^*} is on-path with positive probability. Then strict concavity implies the improvement from a_{i^*} to b_{i^*} is strict at (t, m_{i^*}) . All other coordinates coincide, so b interim weakly dominates a . $\square$

Lemma 10. Fix $k \geq 0$ and $t \in \mathcal{S}^k$. If there exists $i^* \in \{1, \dots, N-1\}$ such that either $t_{i^*} > \tilde{t}_{i^*}^{k+1}$ or $t_{i^*} < \underline{t}_{i^*}^{k+1}$, then t is interim weakly dominated by some $s \in \mathcal{S}^k$ relative to $\mathcal{A}^k$.

Proof. We prove the first case ($t_{i^*} > \tilde{t}_{i^*}^{k+1}$); the second is analogous. Let i^* be the smallest index with $t_{i^*} > \tilde{t}_{i^*}^{k+1}$. Define s by $s_{i^*} := \tilde{t}_{i^*}^{k+1}$ and $s_j := t_j$ for $j \neq i^*$. Only types in $[s_{i^*}, t_{i^*})$ change messages when switching from t to s ; they switch from m_{i^*} to m_{i^*+1} . The verification that $s \in \mathcal{S}^k$ is analogous to the feasibility step in Lemma 9, so omitted. We now argue that s interim weakly dominates t relative to $\mathcal{A}^k$.

(Weakly better against all a .) Fix any $a \in \mathcal{A}^k$. If $a_{i^*} = a_{i^*+1}$ then all switching types are indifferent, so they weakly prefer s to t . Now suppose $a_{i^*} < a_{i^*+1}$. Since $a \leq \tilde{a}^k$, monotonicity of the truncated-root map $\tau(\cdot, \cdot)$ (defined in Section 3.1) implies

$$\tau(a_{i^*}, a_{i^*+1}) \leq \tau(\tilde{a}_{i^*}^k, \tilde{a}_{i^*+1}^k).$$

By definition of the Sender robust best response,

$$\tau(\tilde{a}_{i^*}^k, \tilde{a}_{i^*+1}^k) = \tilde{t}_{i^*}^{k+1},$$

and by construction of the activated upper bound, $\tilde{t}_{i^*}^{k+1} \leq \tilde{t}_{i^*}^{k+1} = s_{i^*}$. Therefore $\tau(a_{i^*}, a_{i^*+1}) \leq s_{i^*}$, so every type $\tau \in [s_{i^*}, t_{i^*})$ strictly prefers a_{i^*+1} to a_{i^*} , and hence strictly prefers s to t against a .

(Strictly better against some a .) By Lemma 7, $\tilde{a}^k$ is strictly increasing, so $\tilde{a}_{i^*}^k < \tilde{a}_{i^*+1}^k$. Applying the previous paragraph to $a = \tilde{a}^k \in \mathcal{A}^k$ implies that all types in the positive-measure set $[s_{i^*}, t_{i^*})$ strictly prefer the higher action and hence strictly prefer s to t against $\tilde{a}^k$. $\square$

C.3 Survival Lemma

Lemma 11. No Receiver strategy in $\mathcal{A}^*$ is interim weakly dominated by another Receiver strategy in $\mathcal{A}^*$, relative to $\mathcal{S}^*$; and analogously for the Sender.

Proof. (Receiver.) Take $a \neq a'$, both in $\mathcal{A}^*$. We will argue that a' does not interim weakly dominate a , relative to $\mathcal{S}^*$. Let $I := \{i \in \{1, \dots, N\} : a_i \neq a'_i\}$. Clearly the dominance cannot hold if $t_{i-1} = t_i$ for all $i \in I$ and all $t \in \mathcal{S}^*$. So assume that is not the case, and choose an $i \in I$ and $t^w \in \mathcal{S}^*$ with $t_{i-1}^w < t_i^w$. Define a continuous path of Sender strategies

$(t^\lambda)_{\lambda \in [0,1]}$ by

$$t^\lambda := \begin{cases} (1 - 2\lambda) \underline{t}^* + 2\lambda t^w, & \lambda \in [0, \frac{1}{2}], \\ (2 - 2\lambda) t^w + (2\lambda - 1) \bar{t}^*, & \lambda \in [\frac{1}{2}, 1]. \end{cases}$$

Since $\mathcal{S}^*$ is convex, we have $t^\lambda \in \mathcal{S}^*$ for all λ . Moreover, $t_i^\lambda - t_{i-1}^\lambda$ is affine in λ on each of the two subintervals $[0, \frac{1}{2}]$ and $[\frac{1}{2}, 1]$, and satisfies $t_i^{1/2} - t_{i-1}^{1/2} = t_i^w - t_{i-1}^w > 0$. Hence $t_i^\lambda - t_{i-1}^\lambda$ is not identically zero on either subinterval, and therefore $t_{i-1}^\lambda < t_i^\lambda$ for all $\lambda \in (0, 1)$ except possibly for at most one value of λ in each subinterval.

Now define the continuous function

$$g(\lambda) := a^R(t_{i-1}^\lambda, t_i^\lambda),$$

with the usual convention $a^R(x, y) = a^R(x)$ if $x = y$. Since $(\underline{t}^*, \underline{a}^*)$ and $(\bar{t}^*, \bar{a}^*)$ are mutual robust best responses, we have $g(0) = \underline{a}_i^*$ and $g(1) = \bar{a}_i^*$. Because $a_i \in [\underline{a}_i^*, \bar{a}_i^*]$, the intermediate value theorem yields $\lambda_0 \in [0, 1]$ such that $g(\lambda_0) = a_i$.

Let $\Pi_i(\lambda)$ be the difference in the Receiver's interim expected payoff from choosing a_i rather than a'_i after message m_i under t^λ . The function Π_i is continuous. Strict concavity implies $\Pi_i(\lambda_0) > 0$ because $a'_i \neq a_i = g(\lambda_0)$, hence $\Pi_i(\lambda) > 0$ for all λ sufficiently close to λ_0 . Choose such a λ with $\Pi_i(\lambda) > 0$ and $t_{i-1}^\lambda < t_i^\lambda$ (so m_i is on path under t^λ). Then a' cannot interim weakly dominate a relative to $\mathcal{S}^*$.

(*Sender.*) Take $t \neq t'$, both in $\mathcal{S}^*$. We will argue that t' does not interim weakly dominate t , relative to $\mathcal{A}^*$. Let $I := \{i \in \{1, \dots, N-1\} : t_i \neq t'_i\}$. Clearly the dominance cannot hold if $a_i = a_{i+1}$ for all $i \in I$ and all $a \in \mathcal{A}^*$. So assume that is not the case, and choose $i \in I$ such that $a_i < a_{i+1}$ for some $a \in \mathcal{A}^*$. Suppose $t'_i > t_i$ (the opposite case is symmetric) and let $i^* := \max\{j \in I : t'_j > t_j\}$. Then $t_{i^*} < t'_{i^*} \leq t_{i^*+1}$, hence $E := [t_{i^*}, t'_{i^*}] \subseteq [t_{i^*}, t_{i^*+1}]$. Therefore every type $\tau \in E$ (a set of positive measure) sends message m_{i^*+1} under t (and so receives action a_{i^*+1}), while under t' its message index is at most i^* (and so the induced action is at most a_{i^*}).

Now consider the path $a^\lambda := (1 - \lambda)\underline{a}^* + \lambda\bar{a}^* \in \mathcal{A}^*$, for $\lambda \in [0, 1]$. Let $c(\lambda)$ be the indifferent type between $a_{i^*}^\lambda$ and $a_{i^*+1}^\lambda$; if no indifferent type exists, then $c(\lambda)$ is 0 if all types prefer the lower action, or 1 otherwise. The function c is continuous; and since $(\underline{t}^*, \underline{a}^*)$ and $(\bar{t}^*, \bar{a}^*)$ are mutual robust best responses, $c(0) = \underline{t}_{i^*}^*$ and $c(1) = \bar{t}_{i^*}^*$. Because $t_{i^*} \in [\underline{t}_{i^*}^*, \bar{t}_{i^*}^*]$, the intermediate value theorem yields $\lambda^* \in [0, 1]$ such that $c(\lambda^*) = t_{i^*}$. Every type $\tau > t_{i^*}$ strictly prefers $a_{i^*+1}^{\lambda^*}$ to $a_{i^*}^{\lambda^*}$, and hence also to any lower action. Therefore, every type in E strictly prefers t to t' , so t' does not interim dominate t relative to $\mathcal{A}^*$. $\square$

C.4 Putting the Pieces Together

Proof of Theorem 2. Part (3) is Lemma 11.

For part (1), Lemma 9 establishes what is needed for the Receiver deletions in (Rk) , and Lemma 10 for the Sender deletions in (Sk) .

For part (2), Lemma 8 implies that for each k ,

$$\mathcal{A}^k = \{a \in \mathcal{A} : \underline{a}^k \leq a \leq \tilde{a}^k\} \quad \text{and} \quad \mathcal{S}^k = \{t \in \mathcal{S} : \underline{t}^k \leq t \leq \tilde{t}^k\}.$$

Taking intersections over k and using $\underline{a}^k \uparrow \underline{a}^*$, $\tilde{a}^k \downarrow \bar{a}^*$, $\underline{t}^k \uparrow \underline{t}^*$, and $\tilde{t}^k \downarrow \bar{t}^*$ (Lemma 2 and Lemma 8) yields

$$\mathcal{A}^\infty = \{a \in \mathcal{A} : \underline{a}^* \leq a \leq \bar{a}^*\} \equiv \mathcal{A}^* \quad \text{and} \quad \mathcal{S}^\infty = \{t \in \mathcal{S} : \underline{t}^* \leq t \leq \bar{t}^*\} \equiv \mathcal{S}^*. \quad \square$$

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